#FABCONSQLCON2026

# FABCON | SQLCON

## Microsoft Fabric COMMUNITY CONFERENCE | Microsoft SQL COMMUNITY CONFERENCE

**ATLANTA** MARCH 16 - 20, 2026

# Microsoft Fabric Capacities

## Essentials for Smarter Operations

Sanjay Raut, Pat Mahoney, Matt Stenzel

Principal PM's - Azure Data CAT

# Acknowledgment of Contributions

**Core Contributors**

Special thanks to Benni, Pat, Chris, Tim, Matthew, Ope, Lukasz, Edgar, Pankaj, Carlos, Matt, Sanjay, and others for their valuable presentations.

**Workshop Inspiration**

This workshop compiles insights from numerous team members' presentations shared at previous conferences, enriching the community's knowledge.

**Gratitude and Support**

The success of this workshop relies on the support and contributions of the CAT and Capacities Engineering Team.

# Meet the Speakers and Proctors

- Speakers
  - Matt Stenzel
  - Sanjay Raut
  - Pat Mahoney

- Proctors
  - Tim Bindas
  - Pratik Patel

# Full-Day Workshop Agenda Timeline

**9:00 AM**

**Session 1 - Capacity Concepts (75 min)**

CU model, bursting, throttling, hands-on lab [1]

**10:30 AM**

**Session 2 - Recovery & Advanced Config (90 min)**

Throttling recovery, advanced settings, licensing [1]

**1:30 PM**

**Session 3 - Monitoring & Events (120 min)**

Metrics App, Real-Time Hub, chargeback demos [1]

**3:30 PM**

**Session 4 - Planning & Optimization (30 min)**

Best practices, scale decisions, workload isolation [1]

**4:00 PM**

**Session 5 - Q&A with Leadership (30 min)**

Open forum with Capacity Engineering Leadership [1]

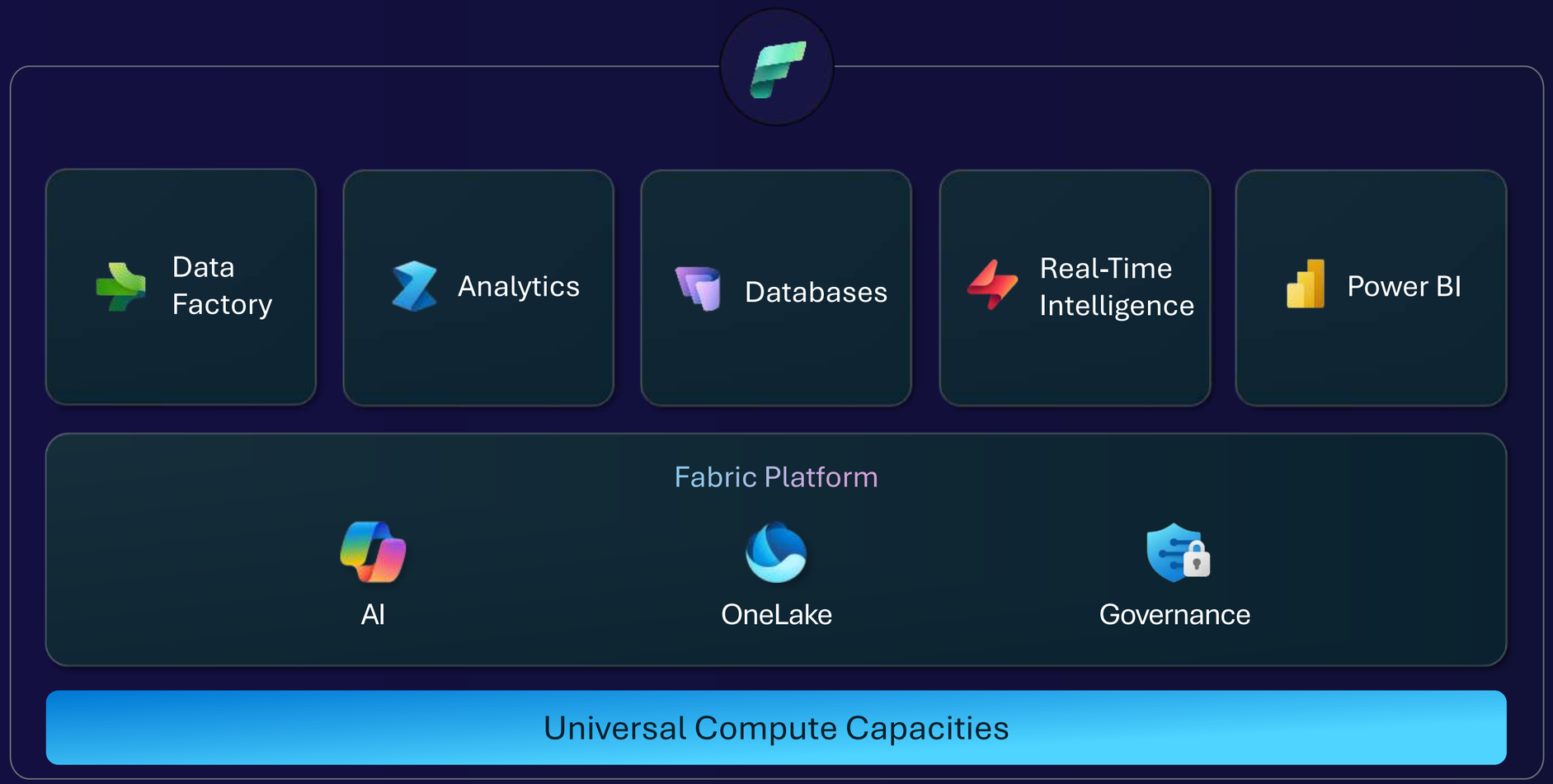
- Structured learning progression: From foundational concepts to advanced topics
- Interactive engagement with hands-on labs, live demonstrations
- WIFI: Info on back of badge
- Bonus - dedicated Q&A sessions towards end of the day to ask further clarifications around real capacity scenarios
- Lunch Break:
  - 12:00 Noon to 1:30 PM

# Session 1 - Fabric Capacities Concepts

## Basics

# Microsoft Fabric

## The unified data and analysis platform for AI transformation

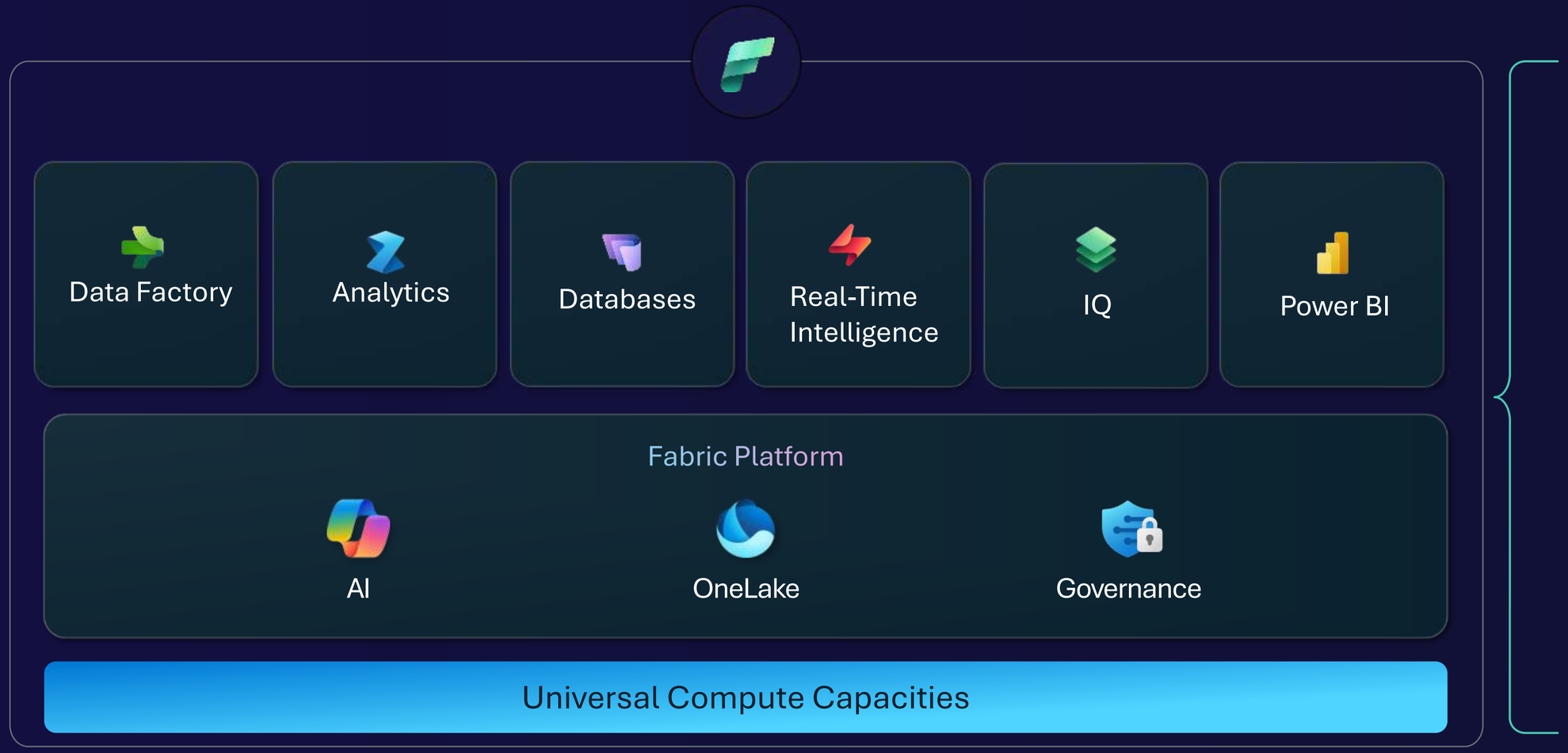



**Single...**

- Onboarding and trials
- Sign-on
- Navigation model
- UX model
- Workspace organization
- Collaboration experience
- Data Lake
- Storage format
- Data copy for all engines
- Security model
- CI/CD
- Monitoring hub
- Data catalog

# Microsoft Fabric

## The unified data and analysis platform for AI transformation

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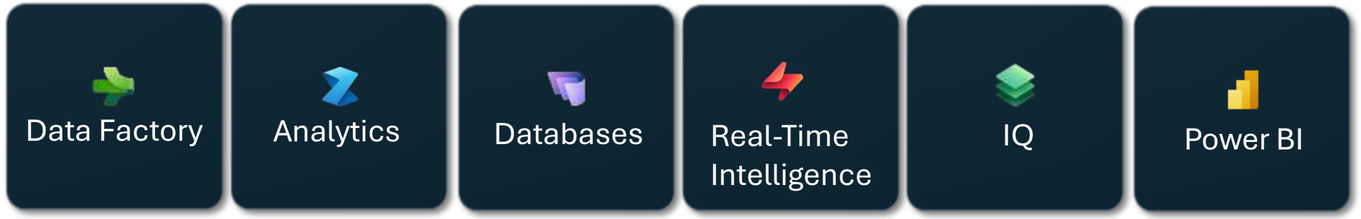
# Capacities are a shared resource

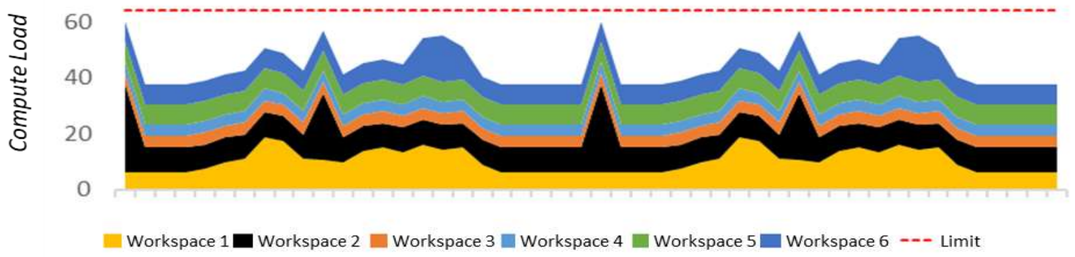
**Shared across workloads**
A single capacity is providing the compute power for all Fabric workloads.

There is no need to allocate compute for each workload separately.



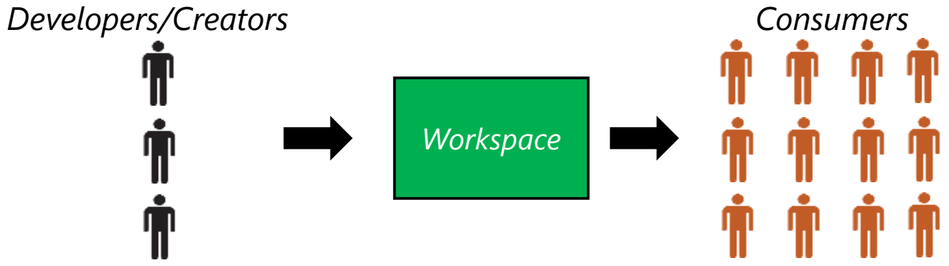

**Shared Across Projects**
A single capacity typically supports dozens of separate projects simultaneously, each managed in its own workspace.

It is rare to have a capacity dedicated to a single project




**Shared across users**
For each project, many developers will share a workspace where collaborative development and consumption at scale is managed.

Each creator can provision any artifact and run any job without the need for any pre-approval or planning



# Fabric Capacity Core Concepts

**Scalable Compute Units**

- **Capacity Unit Seconds (CUs)** are the base compute unit for all Capacities
- Your SKU Size determines the number of CUs you have available
- **Multiple workloads** can use the same capacity at the same time

**Resizeable, Pausable**

- **Fabric SKUs** (F SKUs) enable **added flexibility**
- **Resize** to increase or decrease the SKU to meet your compute needs
- **Pause** and **Resume the capacity**

**Self-Managing with Bursting and Smoothing**

- **Self-Managing with Bursting and Smoothing**
- **Bursting** allows jobs to run at peak performance
- **Smoothing** reduces the impact of spikes in compute

**But as with any resource, you still can push them too far (Throttling)**

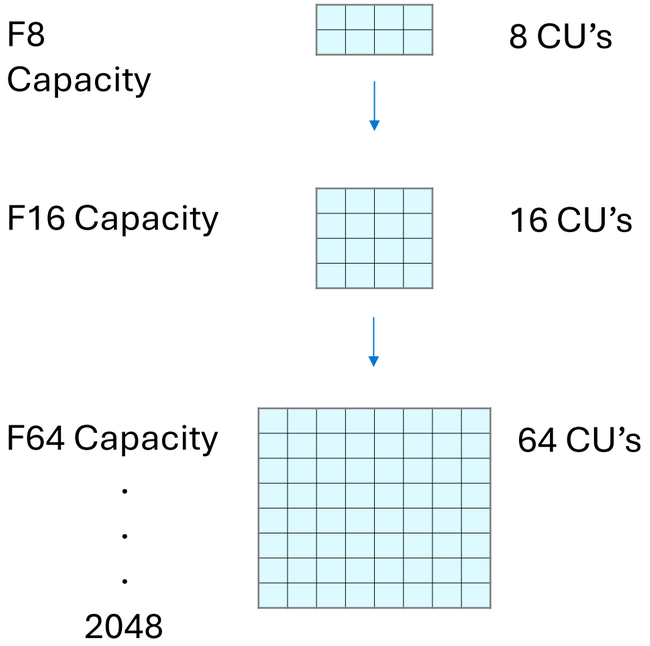
- Capacities offer **built-in resource governance**
- When there's too much smoothed usage, **throttling is applied**
  - **Interactive jobs Delay –** 20s delay, when 10 min > Usage <=60 min
  - **Interactive jobs Rejections –** Rejection, when 60 min > Usage <= 24 hours
  - **Background Rejections –** Rejection, when Usage > 24hrs

# Capacities are flexible building blocks for growth

Capacities can be configured to meet scale, usage and governance requirements while tuning to minimize TCO and performance goals

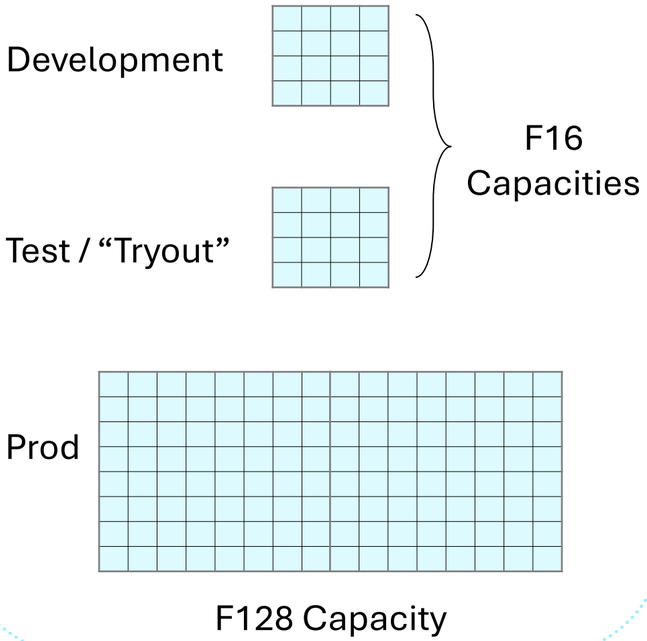
## Scale Vertically

Increased capacity size provides more throughput



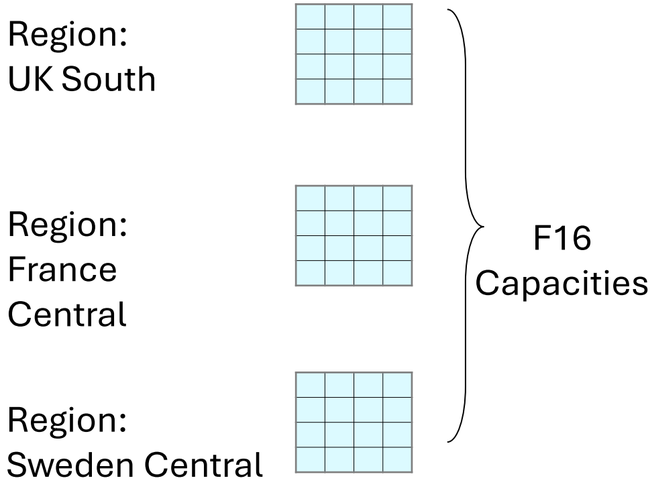

## Scale Horizontally

Scale horizontally using the benefits of modular design for hardened isolation and governance




## Regional Availability

Use different capacities for different regions to support GDPR / Data residency requirements

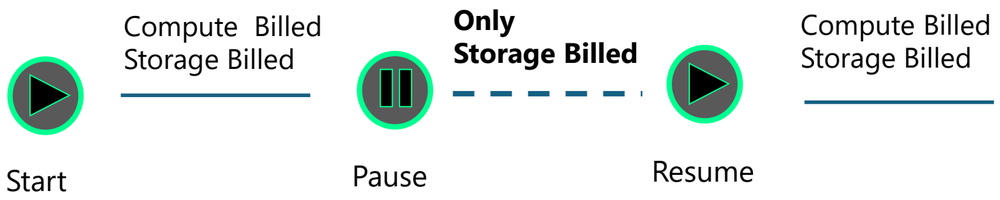


# Provisioning and Deploying Capacities

## Purchased in Azure

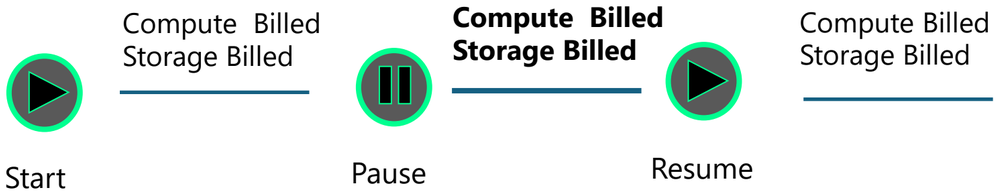
- As a Pay As You Go (**PAYG**) or Reserved Instance (**RI**) resource
- **PAYG:** Capacities are priced at fixed rate

**Pause** and **Resume the capacity**




- The **RI** commitment enjoys a **41% discount**
  - 1-year or 3-year RI
- **RI is scoped** at Billing Account, Subscription, Resource Group or Region level.

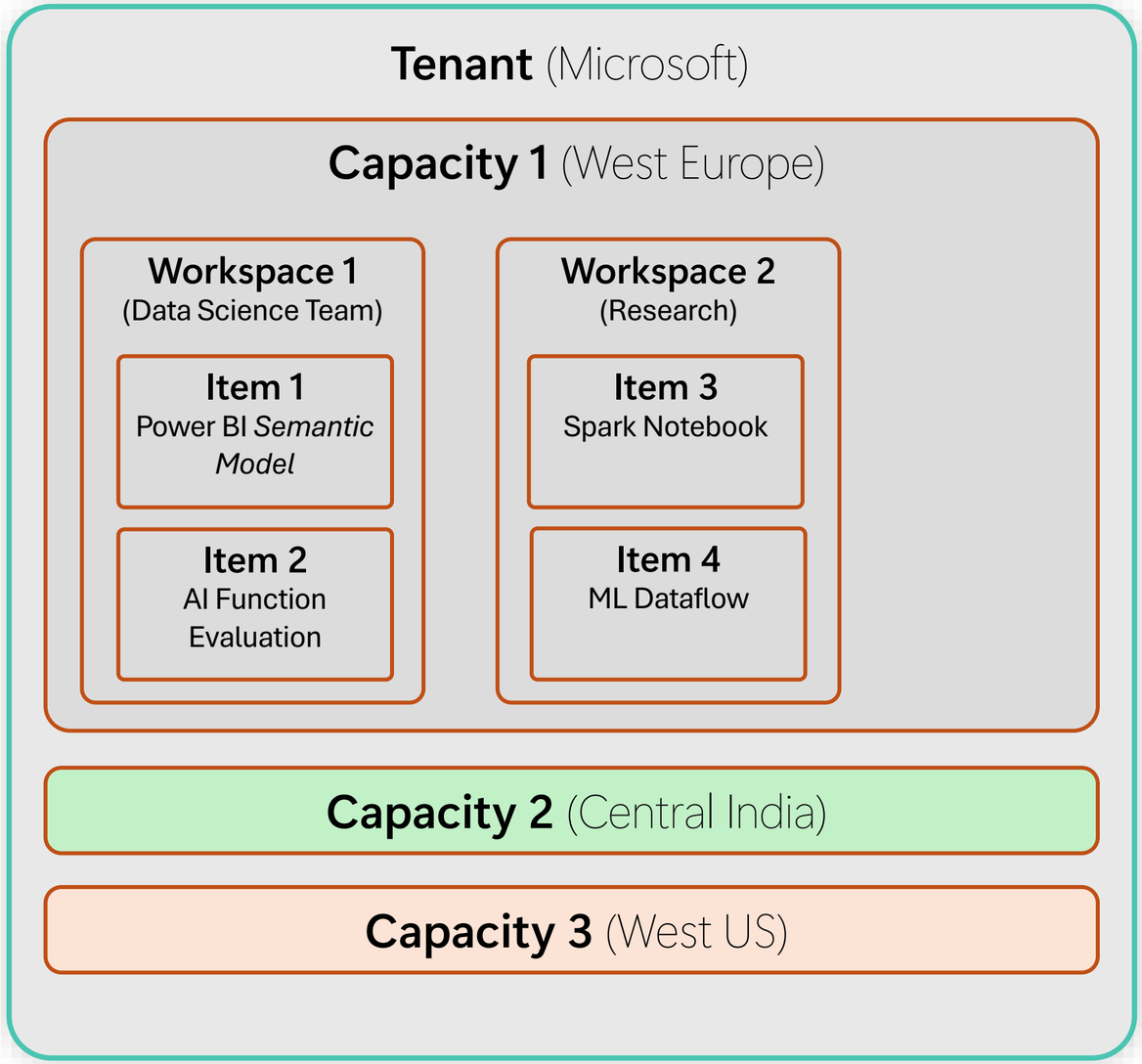
**Pause** and **Resume the capacity**



# Provisioning and Deploying Capacities

## Deployed to Regions

- Each capacity **resides in a specific region of the buyers' choice** where both the data & compute reside
- **Workspaces are assigned to a capacity** that provides the compute and storage for all the workspace artifacts
- Multiple capacities can be purchased, deployed and managed by **different owners** residing in a single tenant allowing each business unit to pay for their own consumption



# Capacity Units, Bursting and Smoothing

## Concepts

# Capacity Unit Seconds

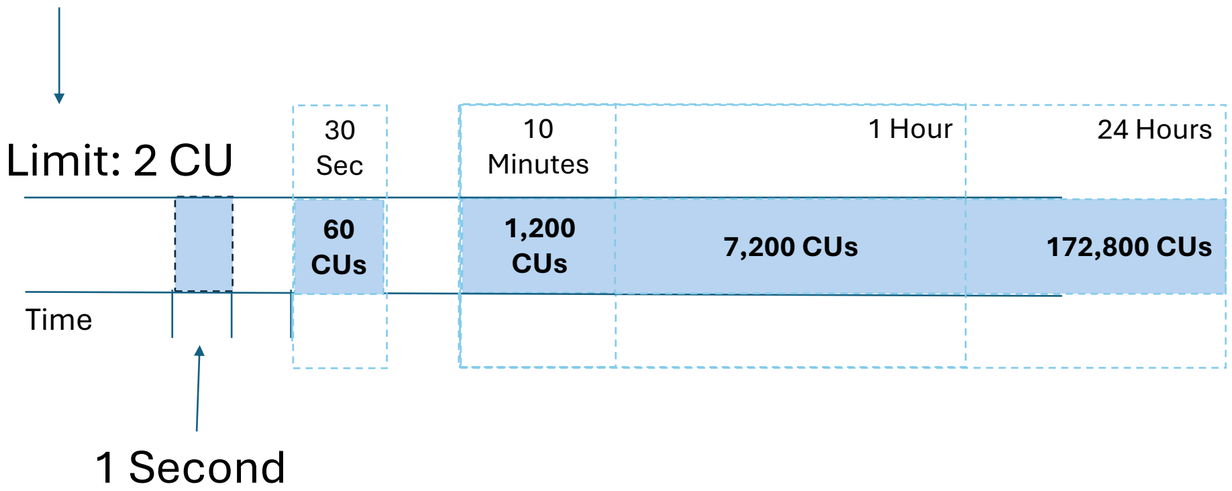
**Capacity Unit Seconds (CUs)** "see-yews" are the base compute unit for all Capacities.

Your Capacity Size determines the number of CUs you have available.

**[# of CU] x [# of seconds] = Available CUs**

[Understand your Fabric capacity Azure bill - Microsoft Fabric | Microsoft Learn]

Example: F2 Capacity has 2 CU per 1 second = **2 CUs**



# Timepoint

## CUs and Timepoint

- **30 seconds** of interval is considered **one timepoint**
- The **more capacity units are provisioned, the more load** the capacity can support
  - Multiply SKU size by **30s** to match platform evaluation in metrics app

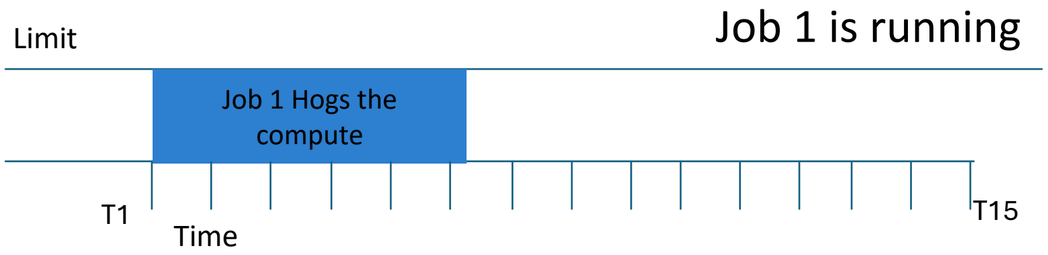
**For example F64**
**[64] x [30] = 1920 CUs**

Universal Compute Capacities SKU Sizing

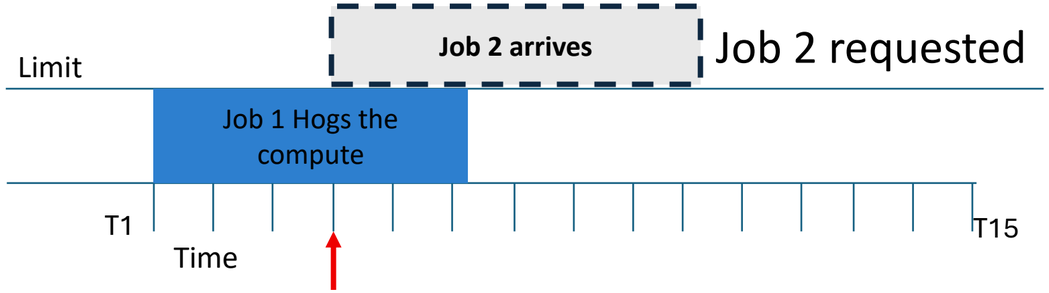
| SKU | Capacity Units (CU) | CU's (per 30s) | Power BI SKU | Power BI V-cores |
|---|---|---|---|---|
| F2 | 2 | 60 | - | 0.25 |
| F4 | 4 | 120 | - | 0.5 |
| F8 | 8 | 240 | A1 | 1 |
| F16 | 16 | 480 | A2 | 2 |
| F32 | 32 | 960 | A3 | 4 |
| F64 | 64 | 1920 | P1 | 8 |
| F128 | 128 | 3840 | P2 | 16 |
| F256 | 256 | 7680 | P3 | 32 |
| F512 | 512 | 15360 | P4 | 64 |
| F1024 | 1024 | 30720 | P5 | 128 |
| F2048 | 2048 | 61440 | - | 256 |

# Bursting and Smoothing

**On Systems** where job 1 is already running at limits



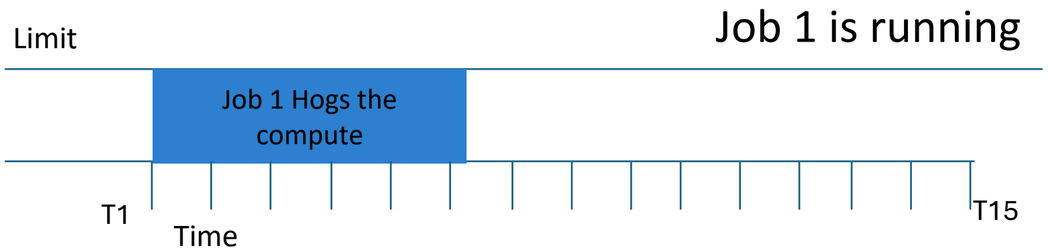

And Job 2 is requested to run at same time



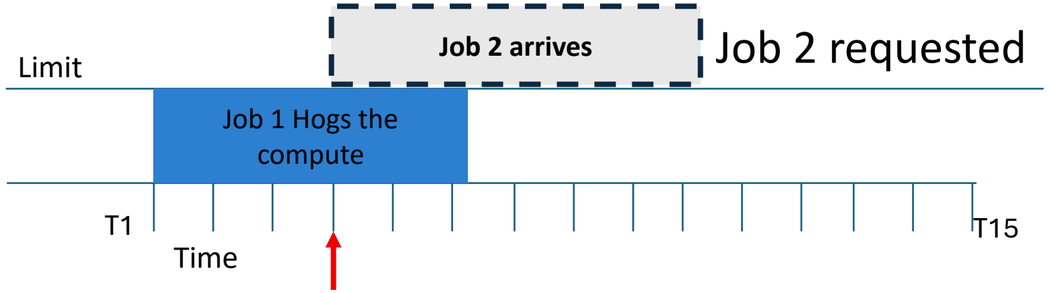

What happens ???

# Bursting and Smoothing

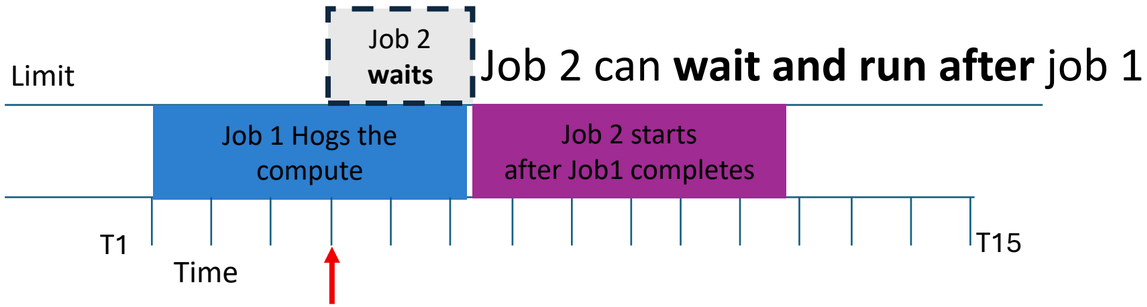
**On Systems** where job 1 is already running at limits

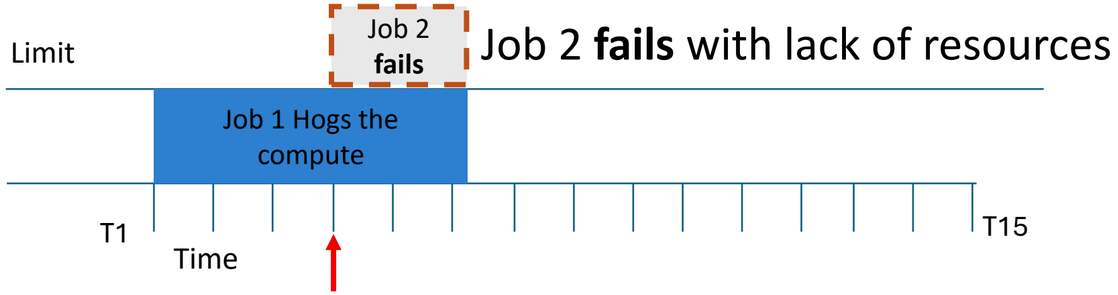


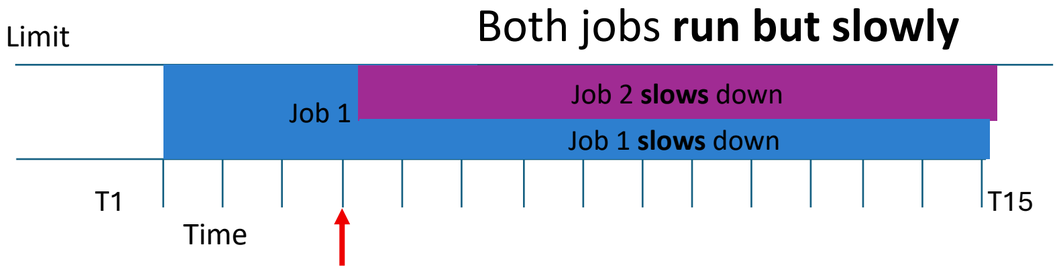

Job 2 is requested to run at same time **What happens???**




**Traditional Systems** stay within a hard compute limit



# Bursting and Smoothing

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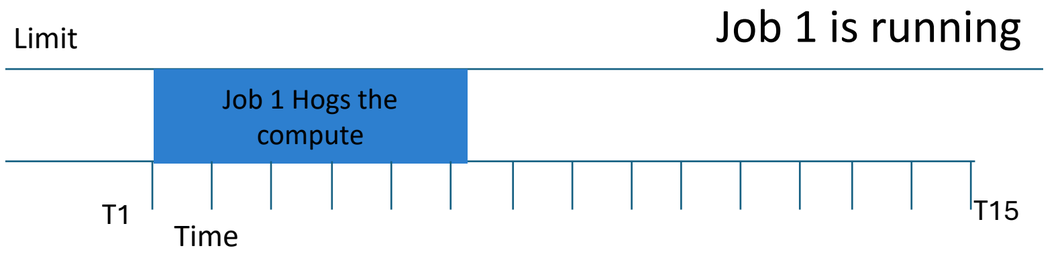


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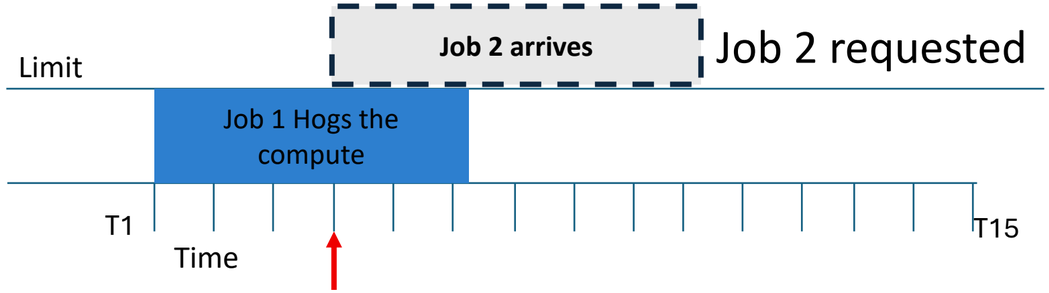
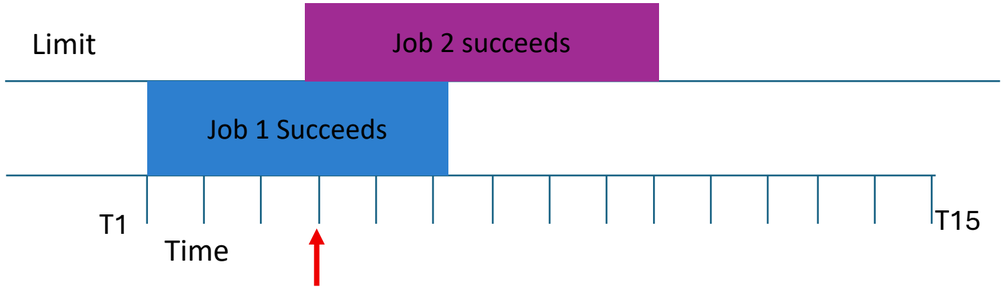
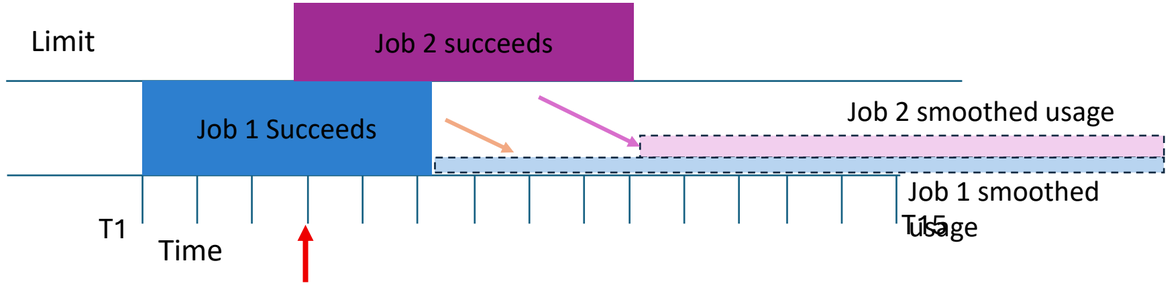



**In Fabric Capacities**
**Bursting & Smoothing helps to solve these problems**

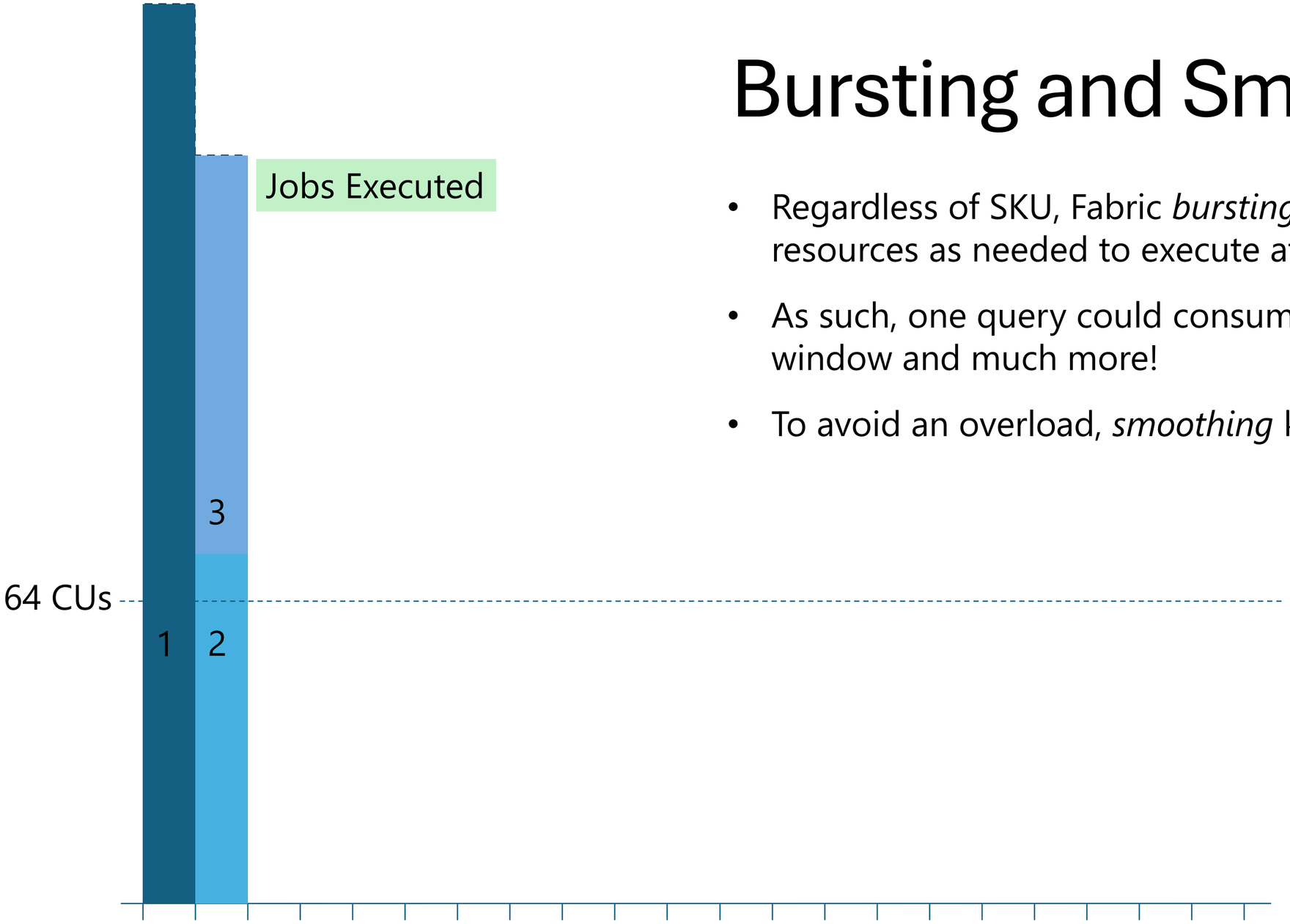
**Bursting** lets both jobs **succeed**, running at **peak performance** to finish **fast**.




**Smoothing** distributes the **cost** of jobs over a longer time-period preventing scheduling issues. Allowing more jobs.

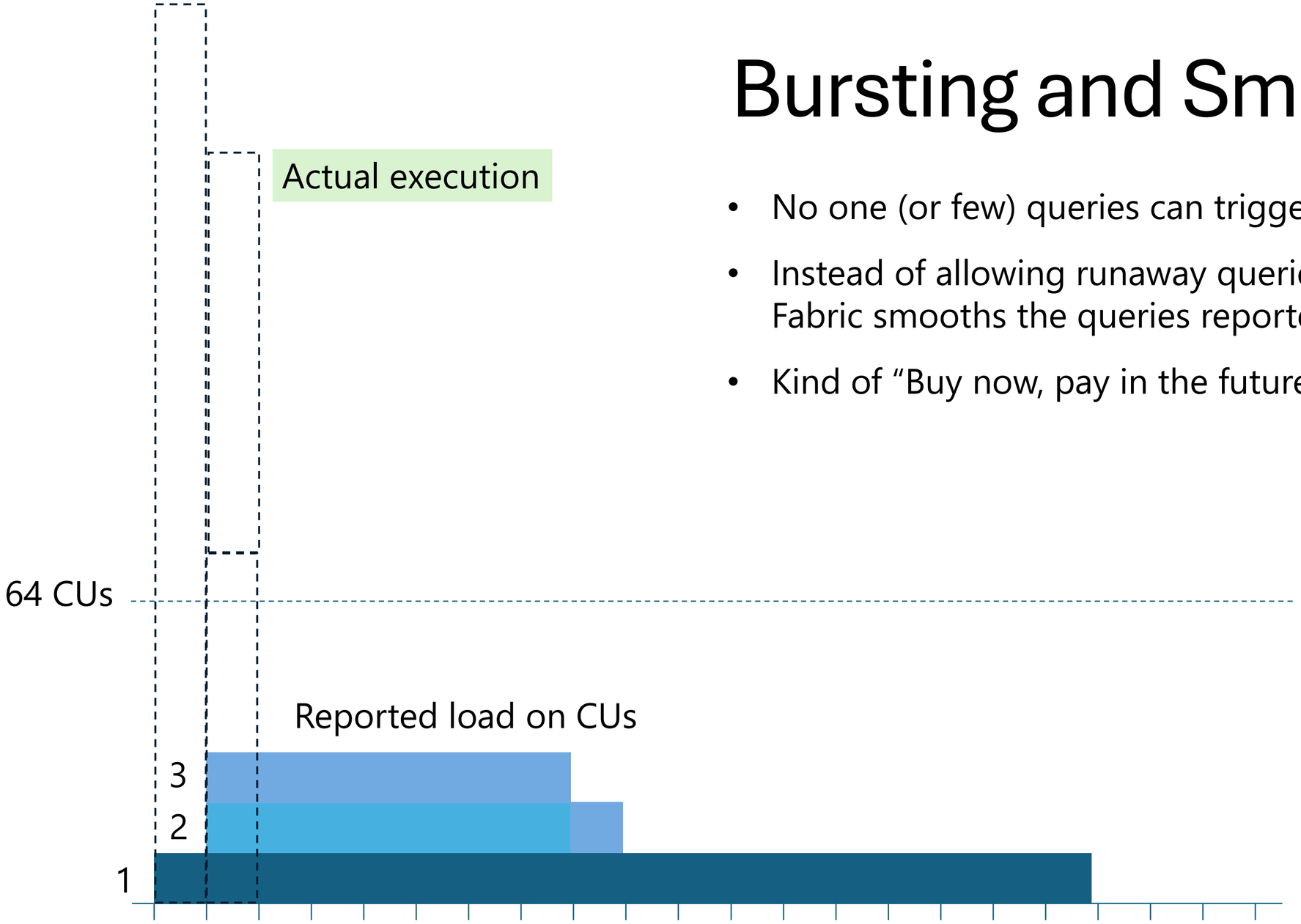


# Bursting and Smoothing



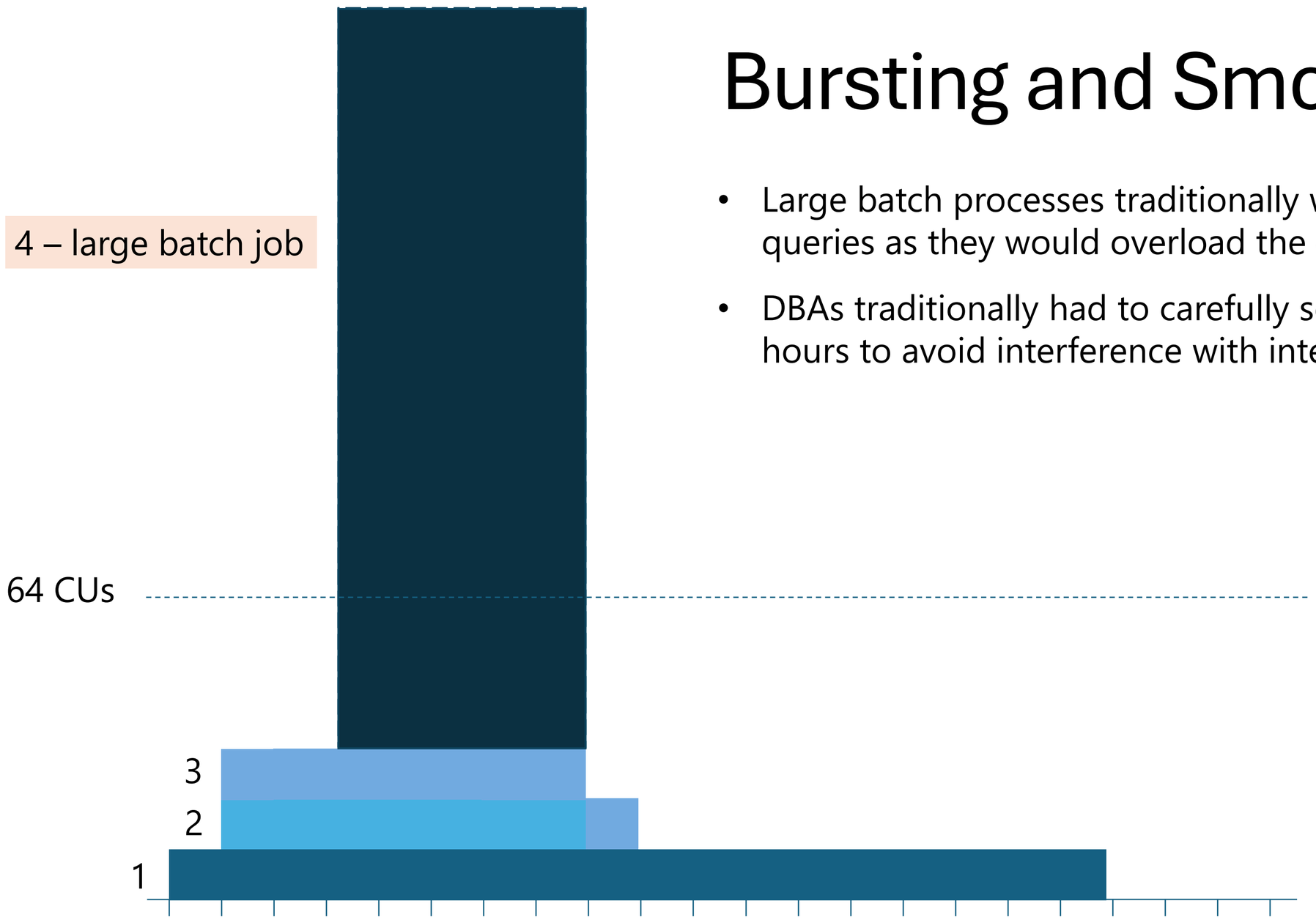

- Regardless of SKU, Fabric *bursting* will automatically allocate resources as needed to execute at maximum performance
- As such, one query could consume all the quota of a single time window and much more!
- To avoid an overload, *smoothing* kicks in

# Bursting and Smoothing




- No one (or few) queries can trigger an overload
- Instead of allowing runaway queries to create a local overload, Fabric smooths the queries reported usage to future time windows
- Kind of “Buy now, pay in the future” installment plan

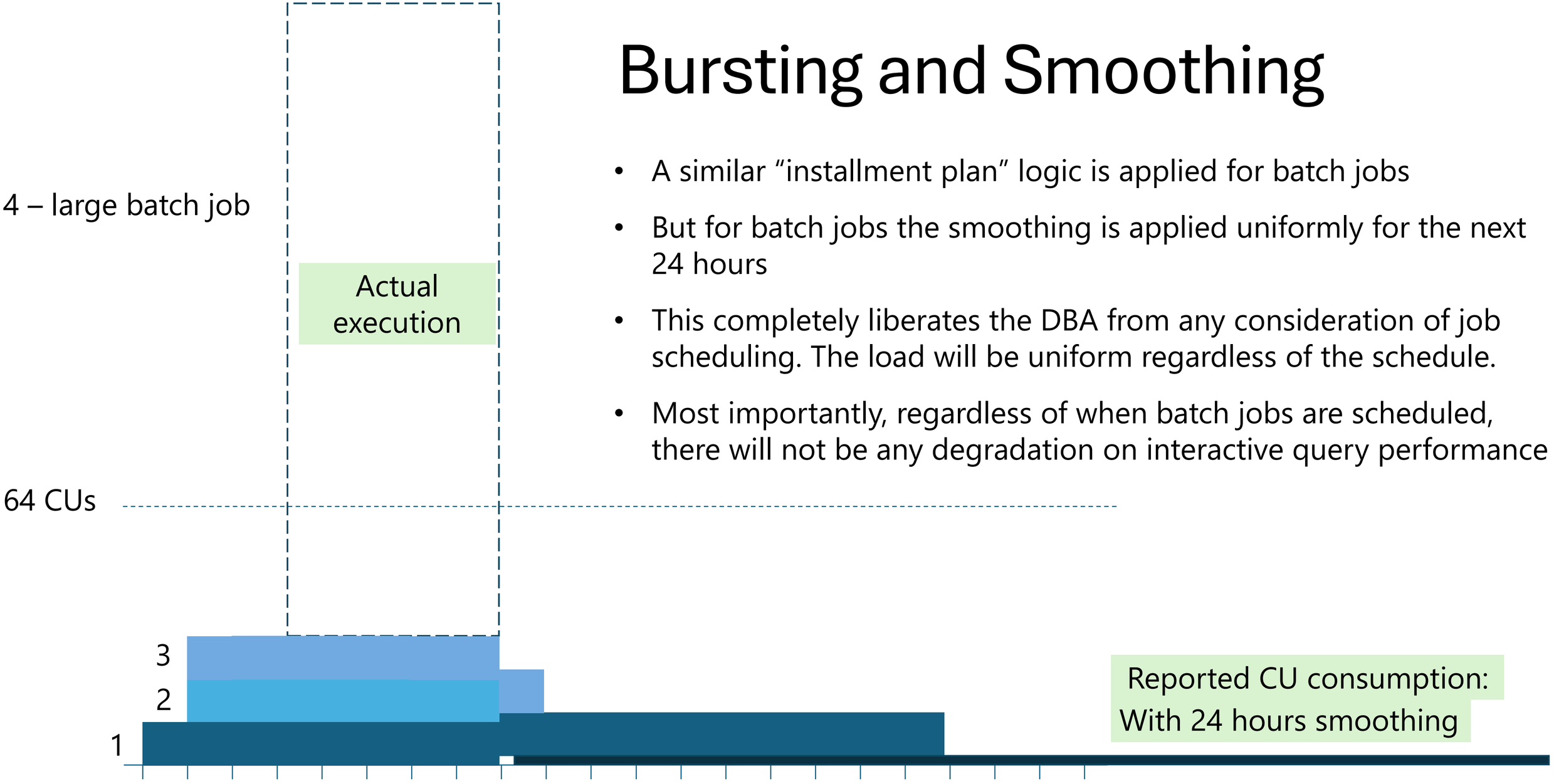
# Bursting and Smoothing




- Large batch processes traditionally were a threat to interactive queries as they would overload the compute resource
- DBAs traditionally had to carefully schedule these jobs to off-hours to avoid interference with interactive user experiences

# Bursting and Smoothing

- A similar “installment plan” logic is applied for batch jobs
- But for batch jobs the smoothing is applied uniformly for the next 24 hours
- This completely liberates the DBA from any consideration of job scheduling. The load will be uniform regardless of the schedule.
- Most importantly, regardless of when batch jobs are scheduled, there will not be any degradation on interactive query performance



# Throttling

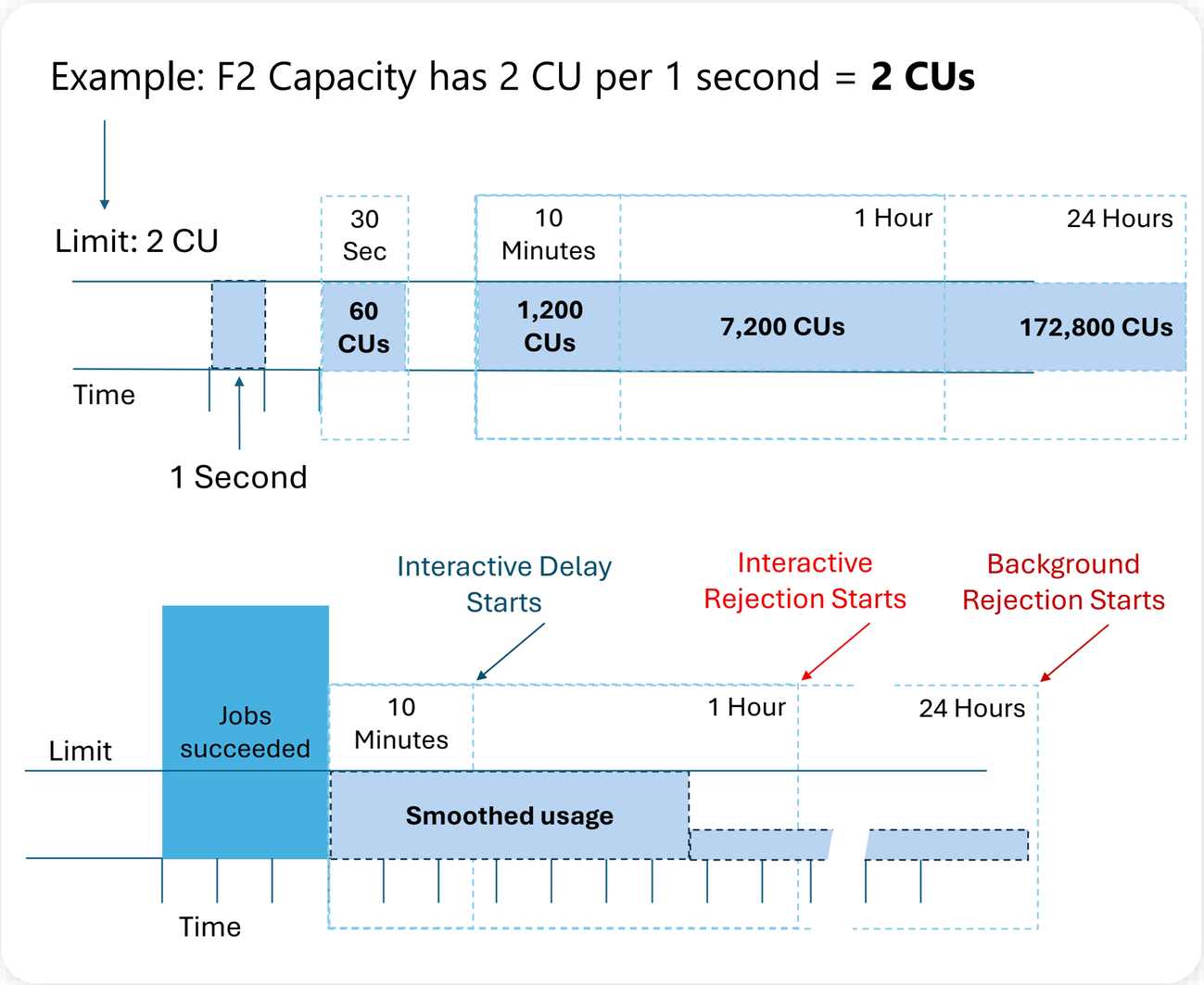
## Concepts

# Throttling

Capacities offer **built-in resource governance**

- A **sustained overuse** will result in **throttling**.
- You won't use your monthly budget in a single day.
- Even overdraft has a limit ☺

**Note:** Users may experience delays, slowness, or failures due to **Workload limits** or **Capacity Throttling**.



# Throttling

**Interactive Delays –**

- Interactive requests will delay their start by **20 seconds**.
- When smoothed usage exceeds future 10 minutes of 100% CUs of the SKU.
- For F2 – future 1,200 CUs are consumed by the smoothed usage

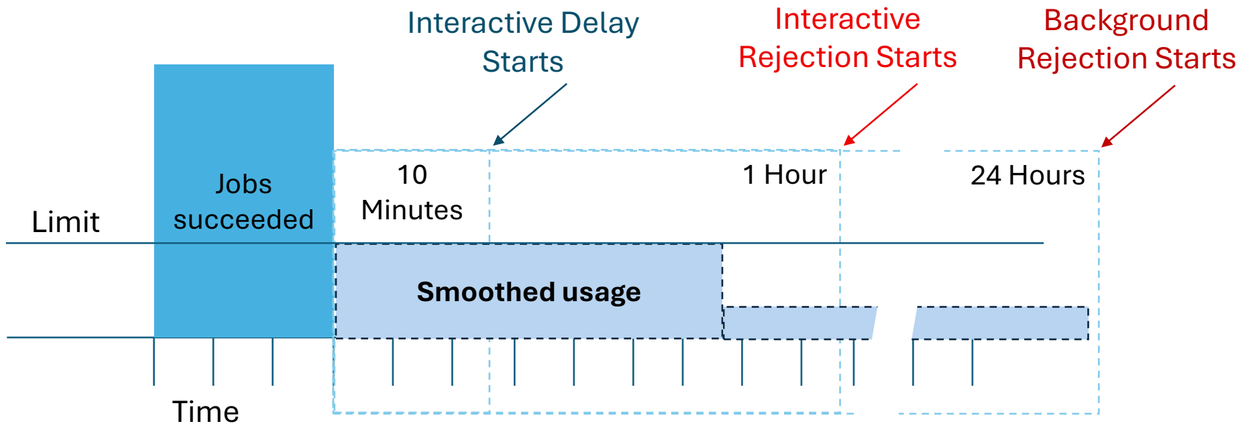
**Interactive Rejections –**

- Interactive requests fail.
- When smoothed usage exceeds future 60 minutes of 100% CUs of the SKU.
- For F2 – future 7,200 CUs are consumed by the smoothed usage

**Background Rejections –**

- All requests fail, including background requests.
- When smoothed usage exceeds future 24 hours of 100% CUs of the SKU.
- For F2 – future 172,800 CUs are consumed by the smoothed usage

[Troubleshooting guide - Diagnose and resolve "capacity limit exceeded" errors]




**How requests are affected by Capacity Throttling**

1. Interactive requests first are **delayed**
2. Then, Interactive requests are **rejected**
3. Then, Background requests are **rejected**

When capacities throttle usage users will see the **CapacityLimitExceeded** error code in the detailed message.

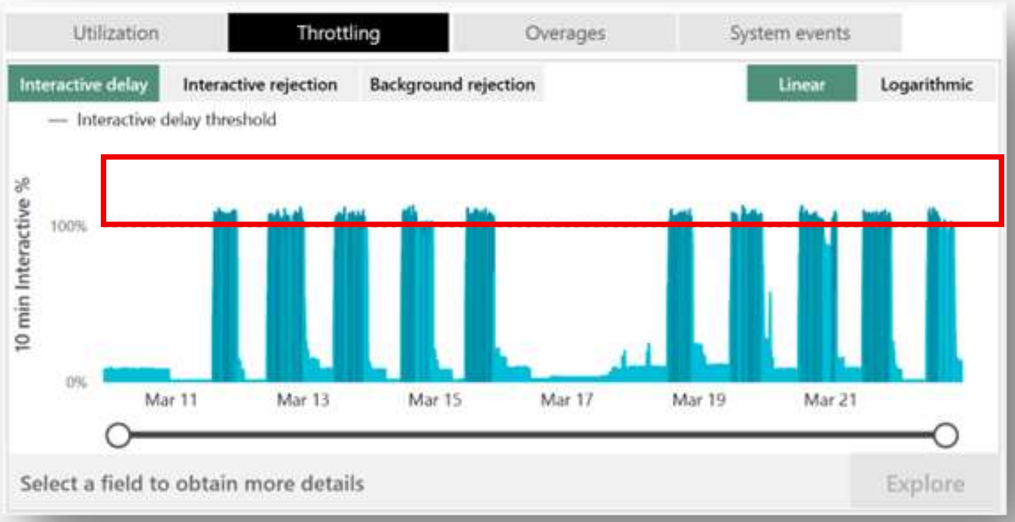
# Capacity Throttling States

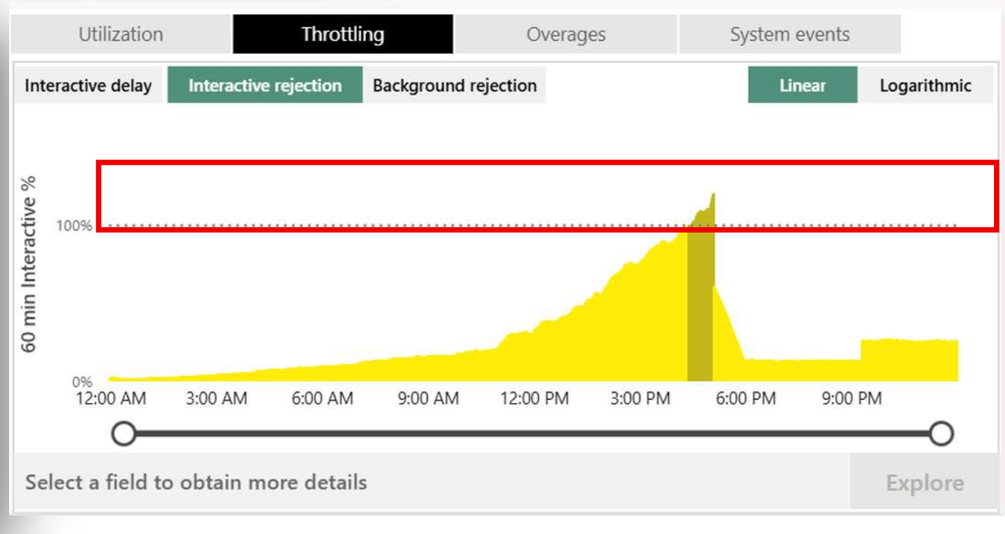
| Smoothed Capacity - Future Use | Platform Policy | Customer Impact |
|---|---|---|
| **<= 10m** | Overage Protection | Jobs can consume 10 minutes of future capacity use without throttling |
| **> 10m → <= 60m** | Interactive Delay | User requested interactive type jobs will be throttled |
| **> 60m → <= 24h** | Interactive Rejection | User requested interactive type jobs will be rejected |
| **> 24h** | Background Rejection | User Scheduled background jobs will be rejected from execution |

# Throttling

Capacity is in throttling state when the throttling tab is showing utilization beyond 100%.
And only for the time period when it is beyond 100% mark on Throttling tab for respective throttling.

Look at Throttling tab for real picture.



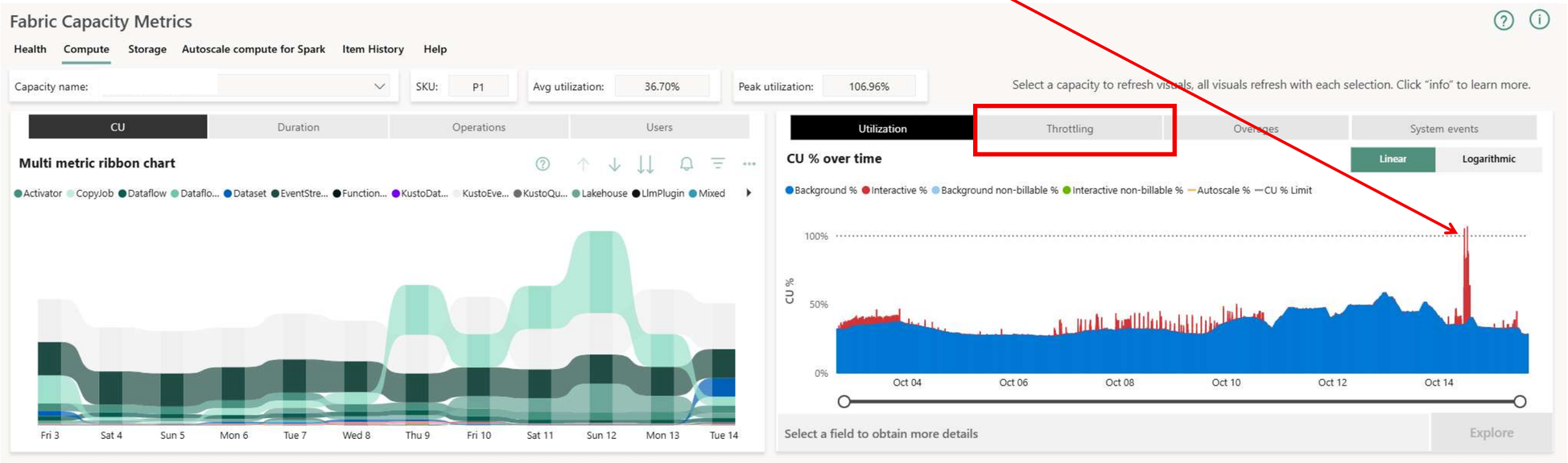

Use [Troubleshooting guide - Diagnose and resolve "capacity limit exceeded" errors] for more guidance on Capacity Troubleshooting

# Throttling

Is my capacity really throttling here

Many times, Admin only look at first tab on compute page

Look at Throttling tab for real picture.



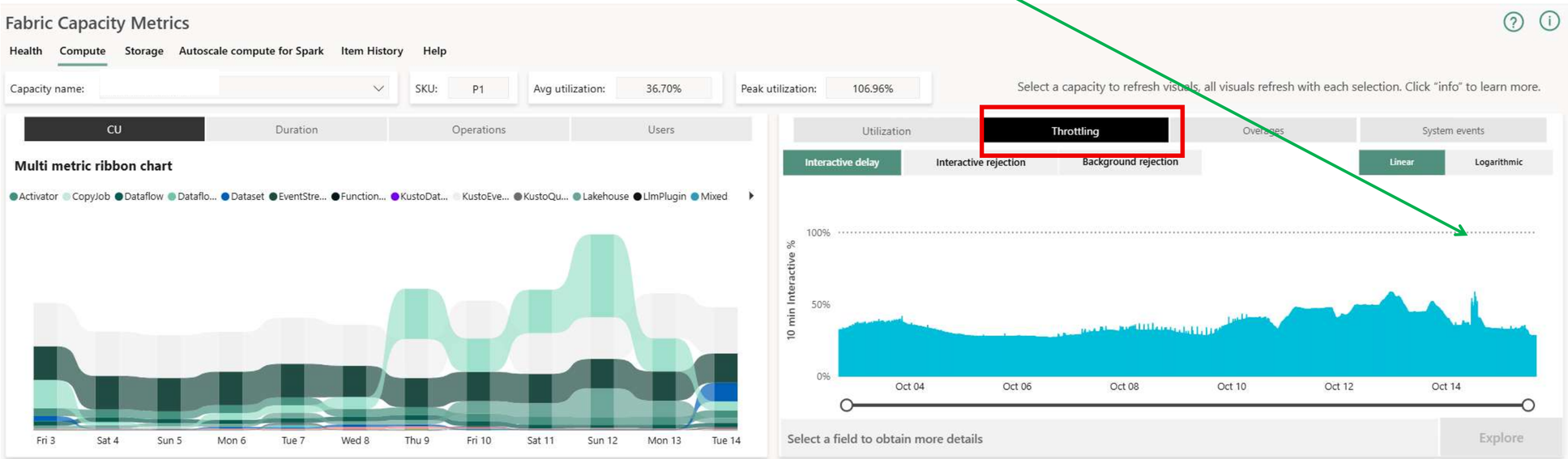

Use [Troubleshooting guide - Diagnose and resolve "capacity limit exceeded" errors]() for more guidance on Capacity Troubleshooting

# Throttling

Is my capacity really throttling here

The answer is “No, It is not.”

Look at Throttling tab for real picture.




Use [Troubleshooting guide - Diagnose and resolve "capacity limit exceeded" errors]() for more guidance on Capacity Troubleshooting

# Sound off.
# The mic is all yours.
# Influence the product roadmap.

Join the Fabric User Panel

Share your feedback directly with our Fabric product group and researchers.

https://aka.ms/JoinFabricUserPanel

Join the SQL User Panel

Influence our SQL roadmap and ensure it meets your real-life needs

https://aka.ms/JoinSQLUserPanel

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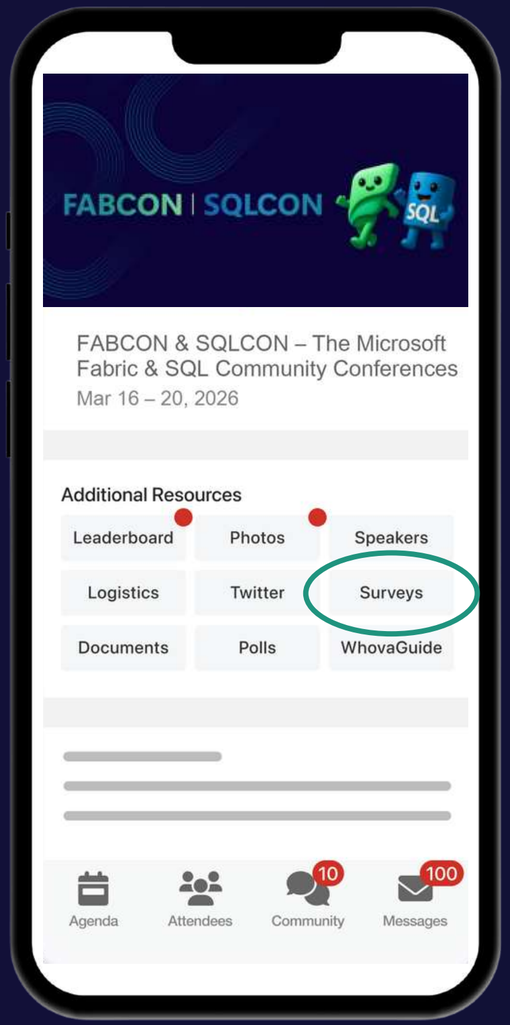

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